JAMA Network Open

Original Investigation | Diabetes and Endocrinology

Glucagon-Like Peptide-1 Receptor Agonists and Fragility Fracture Risk in Type 2 Diabetes

Christopher D. Hamad, MD; Joshua Wiener, BS; Autreen Golzar, BS; Harlene Kaur, BS; Carolyn Henein, BS; Andrew P. Kittredge, BS; Gabriel Su; David C. Kaelber, MD, PhD, MPH; Liana Chan, PhD; Michael R. Yeaman, PhD; John S. Adams, MD; Nicholas M. Bernthal, MD; William L. Sheppard, MD, MPH

Abstract

IMPORTANCE Obesity, type 2 diabetes (T2D), and weight loss are associated with increased fragility fracture risk. Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) are widely prescribed, yet their effects on skeletal outcomes remain uncertain.

OBJECTIVE To evaluate the association between GLP-1 RA initiation and 3-year fragility fracture risk compared with dipeptidyl peptidase-4 inhibitor (DPP-4i) initiation among adults with T2D.

DESIGN, SETTING, AND PARTICIPANTS This comparative effectiveness study using retrospective target trial emulation examined data from the TriNetX Research Network (January 1, 2015, to December 31, 2022), a multicenter US electronic health record database. Adults aged 50 to 90 years with T2D who newly initiated a GLP-1 RA or DPP-4i were followed up for up to 3 years. A separate cohort stratified by T2D status was also analyzed. The database was queried on January 26, 2026, to generate a line-level analytic dataset.

EXPOSURES Initiation of GLP-1 RAs vs DPP-4is.

MAIN OUTCOMES AND MEASURES The primary outcome was incident fragility fracture, defined as fractures after low-energy trauma (eg, fall from standing height). Cohorts were propensity score matched. Time-varying mediation analyses were used to evaluate the contribution of changes in body mass index and hemoglobin A_{1c}.

RESULTS After matching, 133 606 patients (66 803 per group; GLP-1 RA vs DPP-4i: mean [SD] age 63.2 [8.1] vs 63.8 [8.5] years; 35 195 [52.7%] vs 36 098 [54.0%] male) were included. Initiation of GLP-1 RA was associated with lower fragility fracture risk compared with DPP-4i initiation (hazard ratio [HR], 0.79 [95% CI, 0.76-0.83]; absolute risk reduction, 0.79% [95% CI, 0.60%-0.99%]; number needed to treat, 126 [95% CI, 101-168]). The largest risk reductions were observed with vertebral (HR, 0.68 [95% CI, 0.63-0.73]) and hip or femur (HR, 0.70 [95% CI, 0.63-0.79]) fractures. In a sensitivity analysis stratified by diabetes status, fracture risk reduction was observed among patients with T2D (HR, 0.91 [95% CI, 0.88-0.95]) but not among those without T2D (HR, 1.13 [95% CI, 1.04-1.23]; interaction *P* < .001). Mediation analyses showed that the direct association between GLP-1 RA use and lower fracture risk persisted (HR, 0.81 [95% CI, 0.76-0.86]).

CONCLUSIONS AND RELEVANCE This target trial emulation study of adults with T2D found that initiation of a GLP-1 RA was associated with lower 3-year fragility fracture risk compared with initiation of a DPP-4i, independent of changes in body mass index and hemoglobin A_{1c}. Prospective studies are needed to establish causality and define long-term skeletal effects.

JAMA Network Open. 2026;9(7):e2625141. doi:10.1001/jamanetworkopen.2026.25141

Key Points

Question What is the association of glucagon-like peptide-1 receptor agonists (GLP-1 RAs) with fragility fracture risk in patients with type 2 diabetes?

Findings In this comparative effectiveness study using target trial emulation of 133 606 adults with type 2 diabetes mellitus, GLP-1 RA initiation was associated with a statistically significant 21% lower 3-year risk of fragility fracture compared with dipeptidyl peptidase-4 inhibitor initiation.

Meaning These findings suggest that GLP-1 RAs may reduce fragility fracture risk through mechanisms independent of weight loss and glycemic control, highlighting the need for prospective studies to establish causality and define long-term skeletal effects.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

Introduction

Obesity is a growing epidemic in the US. In 2024, obesity affected 40.3% of adults, including 9.4% with severe obesity (body mass index [BMI] >40 [calculated as weight in kilograms divided by height in meters squared]).[1] Concurrently, 14.3% of adults have type 2 diabetes (T2D), and projections suggest that more than 80% of adults in the US may be overweight or obese by 2050.[2,3]

Since the US Food and Drug Administration's approval of exenatide in 2005 for T2D, glucagon-like peptide-1 receptor agonists (GLP-1 RAs) have rapidly expanded in use and indication, now encompassing obesity management, cardiometabolic risk reduction, and treatment of chronic kidney disease in people with T2D.[4-6] According to the 2024 National Health Interview Survey, 26.5% of adults with T2D have used a GLP-1 RA, and per a 2024 Kaiser Family Foundation poll, 22% of adults with obesity have taken one within the past 5 years.[7,8]

Fragility fractures, which result from low-energy trauma (eg, a fall from standing height or lower), are associated with substantial morbidity and mortality, particularly following hip fracture, with reported 1-year all-cause mortality ranging from 12.1% to 25.4% in women and 19.2% to 35.8% in men across multinational cohorts.[9-13] Diabetes, obesity, and rapid weight loss are all concerning in the context of fragility fractures. Adults with T2D, despite their paradoxically higher bone mineral density (BMD), have a 1.79-fold higher risk of hip fracture that may be due to lower bone strength, impaired bone microarchitecture, and accelerated bone loss.[14-16] When controlling for BMD, a U-shaped association exists between BMI and fragility fracture risk, with increased risk among individuals who are underweight or obese.[17] Moreover, rapid or significant weight loss independently increases fracture risk. In a Japanese registry, patients with T2D who lost more than 20% of their body weight had higher hip fracture rates.[18,19] These risks are particularly relevant to users of GLP-1 RAs, as GLP-1 RAs have led to an average of 8% to 21% weight loss across trials and have raised concern that pharmacologic weight loss could increase fragility fracture susceptibility.[20-22]

Despite widespread GLP-1 RA use, their skeletal effects remain insufficiently defined. Preclinical studies have suggested that GLP-1 RAs may promote osteogenic differentiation, inhibit osteoclasts, and signal directly through receptors on osteocytes and mesenchymal precursors.[23] Furthermore, in an ovariectomized rat model, semaglutide preserved bone mineral content and improved bone microarchitecture through β-catenin signaling.[24] However, clinical evidence remains equivocal. The 2025 American Diabetes Association standards of care considered GLP-1 RAs to have a neutral effect on BMD, while select studies and meta-analyses have reported modest fracture risk reductions or GLP-1 RA–specific effects (eg, with exenatide or liraglutide).[25-29]

We therefore evaluated the association between GLP-1 RA initiation and fragility fracture risk using an active comparator target trial emulation restricted to adults with T2D. We conducted sensitivity analyses to also examine GLP-1 RA use among adults with and without T2D to assess generalizability and mediation analyses to assess whether observed associations were mediated by changes in BMI or hemoglobin A_{1c} (HbA_{1c}).

Methods

Data Source and Study Design

We conducted a comparative effectiveness study using the TriNetX Research Network, a federated platform of deidentified electronic health records from US health systems. All data were deidentified under the Health Insurance Portability and Accountability Act privacy rule using expert determination methods pursuant to 45 CFR §164.514(b)(1). Because the investigators used only deidentified data and did not obtain identifiable private information or interact with participants, the study was not considered human participants research under the Common Rule and was exempt from institutional review board oversight and informed consent.

This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) and Transparent Reporting of Observational Studies Emulating a Target Trial (TARGET) reporting guidelines.[30,31] The emulated trial protocol specified eligibility criteria, treatment strategies, treatment assignment, index date, follow-up period, outcome definition, causal contrasts, and statistical analysis plan. The database was queried on January 26, 2026, to generate a line-level analytic dataset.

Cohort Definitions, Exposures, and Outcomes

We identified adults aged 50 to 90 years with T2D per *International Statistical Classification of Disease, Tenth Revision, Clinical Modification* (*ICD-10-CM*) code E11 or HbA_{1c} of at least 6.5% (48 mmol/mol) in the 365 days before index who newly initiated a GLP-1 RA or dipeptidyl peptidase-4 inhibitor (DPP-4i) between January 1, 2015, and December 31, 2022. Patients initiating GLP-1 RAs comprised the treatment group, and DPP-4i initiators comprised the active comparator group, reflecting clinically comparable second-line therapy. Although the GLP-1 RA exposure definition encompassed all formulations, including those with weight management indications, all patients in the primary cohort met T2D inclusion criteria at baseline (eTable 1 in Supplement 1). Initiation followed a new-user design, with the index defined as the first dispensing and requiring at least 4 administrations to ensure continuous use. Eligible patients had continuous health care engagement (≥1 BMI measurement, ≥1 HbA_{1c} test, and ≥4 ambulatory visits) and no fragility fracture in the prior 365 days. Baseline use of other glucose-lowering agents (metformin, sodium-glucose cotransporter 2 inhibitors, thiazolidinediones, insulin, or sulfonylureas) were adjusted for in propensity score models. The primary outcome was time to first fragility fracture within 3 years. A sensitivity analysis excluding the aforementioned agents was performed to address potential confounding by medication-related hypoglycemia and fall risk. Follow-up began the day after index and continued until fracture, death, loss to follow-up (>365-day care gap), or administrative censoring at 3 years. High-energy trauma fractures were excluded, as were patients with osteoporosis, osteopenia, vitamin D deficiency, advanced chronic kidney disease (stage IV-V), end-stage kidney disease, kidney osteodystrophy, or type 1 diabetes (eTable 1 in Supplement 1).

Covariates and Propensity Score Matching

We prespecified 40 baseline covariates measured in the 365 days before index, including demographics, BMI, HbA_{1c}, glycemic medication use, and all Charlson Comorbidity Index components. Race and ethnicity were recorded in the structured electronic health record fields of the contributing health care organizations and standardized into the TriNetX network's categories. Due to the deidentified and aggregated nature of data across institutions, the method of ascertainment at each site was not captured. Race categories included American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or other Pacific Islander, White, other race, and unknown. Ethnicity was classified as Hispanic or Latino, not Hispanic or Latino, or unknown. Race and ethnicity were included as baseline covariates because they may be associated with diabetes prevalence and severity, medication access and prescribing patterns, fracture risk, and health care use. Functional and fracture risks were also captured using fall-related *ICD-10-CM* codes and the Hospital Frailty Risk Score, a validated index derived from 109 weighted *ICD-10-CM* codes (eTable 1 in Supplement 1).[32] Propensity scores were estimated using multivariable logistic regression and applied in 1:1 nearest-neighbor matching (0.2-SD caliper), with exact matching on age (5-year bands) and calendar year (2-year bands). Covariate balance was assessed using standardized mean differences (SMDs), with an SMD less than 0.10 indicating adequate balance.

Statistical Analysis

Primary Group and Subgroup Analyses

Analyses were performed using R, version 4.5.1 (R Foundation for Statistical Computing). In the propensity score–matched (PSM) cohort, hazard ratios (HRs) and 95% CIs for time to first fragility fracture were estimated using Cox proportional hazards models stratified on matched pairs under intention-to-treat and per-protocol frameworks. Intention-to-treat analyses were censored at persistent health care gaps (>365 days), whereas per-protocol analyses were additionally censored at treatment crossover or discontinuation (>90-day exposure gap). Prespecified subgroup analyses evaluated effect modification by age, sex, frailty, and anatomic site, with robustness assessed through adherence-restricted and medication-restricted sensitivity analyses. To assess differential effects by diabetes status, separate PSM cohorts were constructed independently of the primary analysis. Adults aged 50 to 90 years who initiated a GLP-1 RA were matched 1:1 with noninitiators identified using a pseudo-index date defined by the first paired BMI and HbA_{1c} measurement (eTable 2 in Supplement 1). Propensity score estimation was repeated within each stratum, with a 30-day landmark applied to mitigate differential informed presence bias from asymmetric anchor definitions. Additional sensitivity analyses included augmented inverse probability weighting, inverse probability of treatment weighting with Cox regression, and Fine-Gray models to account for the competing risk of death. Exploratory subgroup and site-specific results are reported as HRs with 95% CIs without multiplicity adjustment. For each prespecified subgroup, propensity scores were reestimated within the restricted sample, and matching was repeated to ensure covariate balance (all SMDs <0.10). Effect modification was assessed using Wald tests of interaction terms within Cox models. Statistical significance was defined as a 2-sided $P < .05$.

Mediation Analysis

We performed a time-varying causal mediation analysis with quarterly follow-up, jointly modeling BMI and HbA_{1c} as mediators until fracture or censoring. Total effects represented the overall association between GLP-1 RA use and fracture risk, indirect effects reflected mediation through longitudinal changes in BMI and HbA_{1c}, and direct effects captured the remaining association independent of these mediators. Effects were estimated using the product-of-coefficients method, with matching weights preserving the average treatment effect on the treated estimand.

E-Value Calculation

To assess robustness to unmeasured confounding, we calculated E-values for the point estimate and upper confidence bound of the primary HR. This method represents the minimum strength of association an unmeasured confounder would need to have with both exposure and outcome to fully explain the observed association.

Results

Study Population and Baseline Characteristics

After eligibility criteria and propensity score matching (**Figure 1**), the analytic cohort included 66 803 matched pairs (133 606 total patients) (GLP-1 RA vs DPP-4i initiators: mean [SD] age, 63.2 [8.1] vs 63.8 [8.5] years; 31 608 [47.3%] vs 30 705 [46.0%] female and 35 195 [52.7%] vs 36 098 [54.0%] male; 3756 [5.6%] vs 4200 [6.3%] identifying as Asian, 10 789 [16.2%] vs 10 953 [15.9%] as Black, and 39 420 [59.0%] vs 38 665 [57.9%] as White race) with adequate covariate balance (all SMDs <0.10) (**Table**). Baseline BMI and HbA_{1c} were similar between groups (mean, 33.0 [6.1] vs 32.9 [6.2] for GLP-1 RA and DPP-4i initiators, respectively; SMD = 0.03). Among GLP-1 RA initiators, dulaglutide, semaglutide, and liraglutide accounted for 91.0% of index prescriptions (eTable 3 in Supplement 1). Mean (SD) follow-up was 2.84 (0.56) years for GLP-1 RA initiators and 2.75 (0.70) years for DPP-4i initiators. Treatment persistence (year 3: 50 733 of 59 630 [85.1%] for DPP-4is vs 53 950 of 62 151 [86.8%] GLP-1 RAs), deaths (3153 [4.7%] for DPP-4is and 1702 [2.5%] for GLP-1RAs), and administrative health care gaps (3155 [4.7%] for DPP-4i and 2221 [3.3%] for GLP-1 RAs) were similar between groups (detailed cohort disposition provided in eTable 4 in Supplement 1).

Primary Analysis: Overall Fragility Fracture Risk

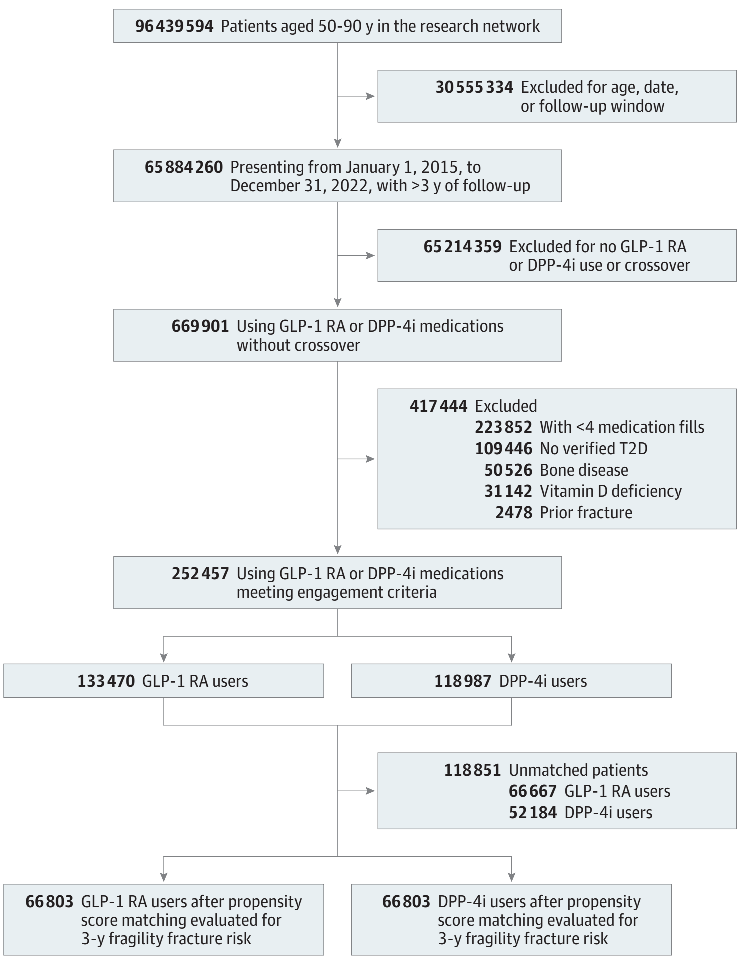
Over 3 years of follow-up, fragility fractures occurred in 2030 GLP-1 RA and 2484 DPP-4i initiators. Initiation of GLP-1 RA was associated with lower fracture risk (HR, 0.79 [95% CI, 0.76-0.83]), corresponding to an absolute risk reduction of 0.79% (95% CI, 0.60%-0.99%) and a number needed to treat of 126 (95% CI, 101-168). Risk reduction was significant in year 1 (HR, 0.72 [95% CI, 0.67-0.78]) and year 2 (HR, 0.77 [95% CI, 0.71-0.83]), with attenuation by year 3 (HR, 0.94 [95% CI, 0.86-1.02]) (**Figure 2**). Results were consistent across sensitivity analyses (Figure 2; eFigure in Supplement 1).

Subgroup Stratification

Fracture risk reduction was consistent across age groups, between sexes, and across all frailty strata (Figure 2). In the site-specific fracture analysis, protection varied by fracture site, with the largest risk reductions associated with vertebral (HR, 0.68 [95% CI, 0.63-0.73]), hip or femur (HR, 0.70 [95% CI, 0.63-0.79]), and rib (HR, 0.83 [95% CI, 0.77-0.91]) fractures and no associations for distal radius or ulna (HR, 1.03 [95% CI, 0.90-1.18]) or proximal humerus (HR, 0.93 [95% CI, 0.82-1.06]) fractures (Figure 2).

Among 67 458 patients with longitudinal BMI and HbA_{1c} data included in the mediation analysis, findings showed that the protective association between GLP-1 RA use and fracture risk (HR, 0.81; 95% CI, 0.76-0.86) was driven primarily by a direct association. In the intention-to-treat cohort, mean (SD) HbA_{1c} decreased by 0.7% (1.4%) with GLP-1 RAs vs 0.5% (1.4%) with DPP-4is, and mean BMI decreased by 0.7 (1.9) vs 0.4 (1.8). In contrast, cumulative BMI loss was associated with increased fracture risk (HR, 1.02 [95% CI, 1.01-1.02]), whereas cumulative HbA_{1c} reduction was associated with a modest risk reduction (HR, 0.99 [95% CI, 0.98-0.99]). The combined indirect

Figure 1. Flowchart of the Cohort Creation



DPP-4i indicates dipeptidyl peptidase-4 inhibitor; GLP-1 RA, glucagon-like peptide-1 receptor agonist; T2D, type 2 diabetes.

Table. Baseline Demographics and Clinical Characteristics After Propensity Score Matching Among Patients Initiating GLP-1 RAs or DPP-4is

Characteristic	Patients, No. (%)					
	Before matching			After matching		
	GLP-1 RA	DPP-4i	SMD[a]	GLP-1 RA	DPP-4i	SMD[a]
No. of patients	133 470	118 987		66 803	66 803	
Age, mean (SD), y	61.6 (7.7)	66.2 (9.0)	−0.54	63.2 (8.1)	63.8 (8.5)	−0.07
Sex						
Female	66 809 (50.1)	53 165 (44.7)	0.11	31 608 (47.3)	30 705 (46.0)	0.03
Male	66 661 (49.9)	65 822 (55.3)	−0.11	35 195 (52.7)	36 098 (54.0)	−0.03
Race						
Asian	4054 (3.0)	18 687 (15.7)	−0.45	3756 (5.6)	4200 (6.3)	−0.03
Black	23 443 (17.6)	13 120 (11.0)	0.19	10 789 (16.2)	10 593 (15.9)	0.01
White	90 139 (67.5)	46 719 (39.3)	0.59	39 420 (59.0)	38 665 (57.9)	0.02
BMI, mean (SD)	35.3 (6.4)	30.7 (6.5)	0.72	33.0 (6.1)	32.9 (6.2)	0.03
HbA_{1c}, mean (SD), %[b]	8.53 (2.0)	8.23 (1.8)	0.16	8.48 (1.9)	8.40 (1.9)	0.04
Charlson Comorbidity Index, mean (SD)[c]	2.35 (1.8)	2.46 (2.0)	−0.06	2.43 (1.9)	2.40 (1.9)	0.02
HFRS						
Low (<5)	96 781 (72.5)	90 872 (76.4)	−0.09	48 729 (72.9)	49 286 (73.8)	−0.02
Intermediate (5-15)	30 670 (23.0)	22 211 (18.7)	0.11	14 702 (22.0)	13 991 (20.9)	0.03
High (>15)	6019 (4.5)	5904 (5.0)	−0.02	3372 (5.1)	3526 (5.3)	−0.01
CKD stage						
1	559 (0.4)	605 (0.5)	−0.01	320 (0.5)	313 (0.5)	<0.01
2	1941 (1.5)	1966 (1.7)	−0.02	1072 (1.6)	1026 (1.5)	0.01
3	9988 (7.5)	10 257 (8.6)	−0.04	5632 (8.4)	5735 (8.6)	−0.01
2-y Fall history	3142 (2.4)	5730 (4.8)	−0.13	1881 (2.8)	2057 (3.1)	−0.02
Bone-active medication	929 (0.7)	1334 (1.1)	−0.05	590 (0.9)	621 (0.9)	−0.01
Glucocorticoid use	1251 (0.9)	740 (0.6)	0.04	594 (0.9)	550 (0.8)	0.01
Hypoglycemia	3702 (2.8)	1976 (1.7)	0.08	1704 (2.6)	1411 (2.1)	0.03
Hormone replacement therapy	2946 (2.2)	1269 (1.1)	0.09	1282 (1.9)	1025 (1.5)	0.03
Antidiabetes medications						
Metformin	78 574 (58.9)	81 798 (68.8)	−0.21	41 667 (62.4)	42 909 (64.2)	−0.04
Insulin	51 484 (38.6)	34 758 (29.2)	0.20	25 158 (37.7)	23 090 (34.6)	0.06
SGLT2 inhibitor	25 588 (19.2)	13 808 (11.6)	0.21	11 901 (17.8)	10 281 (15.4)	0.07
Sulfonylurea	30 275 (22.7)	36 254 (30.5)	−0.18	17 925 (26.8)	18 822 (28.2)	−0.03
Thiazolidinedione	6061 (4.5)	8103 (6.8)	−0.10	3664 (5.5)	3823 (5.7)	−0.01
Comorbidities						
Myocardial infarction	8678 (6.5)	7500 (6.3)	0.01	4625 (6.9)	4598 (6.9)	<0.01
Heart failure	11 087 (8.3)	10 149 (8.5)	−0.01	5940 (8.9)	6101 (9.1)	−0.01
Peripheral vascular disease	643 (0.5)	548 (0.5)	<0.01	350 (0.5)	369 (0.6)	< −0.01
Cerebrovascular disease	10 483 (7.9)	14 691 (12.4)	−0.15	6361 (9.5)	6488 (9.7)	−0.01
Dementia	1027 (0.8)	3437 (2.9)	−0.16	810 (1.2)	887 (1.3)	−0.01
COPD	27 904 (20.9)	16 880 (14.2)	0.18	12 556 (18.8)	11 662 (17.5)	0.04
Rheumatic disease	2350 (1.8)	1386 (1.2)	0.05	1082 (1.6)	946 (1.4)	0.02
Peptic ulcer disease	2229 (1.7)	6725 (5.7)	−0.21	1526 (2.3)	1481 (2.2)	0.01
Mild liver disease	12 871 (9.6)	8248 (6.9)	0.10	5446 (8.2)	4790 (7.2)	0.04
Severe liver disease	916 (0.7)	1026 (0.9)	−0.02	537 (0.8)	540 (0.8)	< −0.01
Diabetes with complications	46 394 (34.8)	35 386 (29.7)	0.11	23 383 (35.0)	21 500 (32.2)	0.06
Kidney disease	14 138 (10.6)	14 808 (12.5)	−0.06	8005 (12.0)	8098 (12.12)	< −0.01
Nonmetastatic cancer	12 390 (9.3)	13 660 (11.5)	−0.07	6797 (10.2)	6940 (10.39)	−0.01
Metastatic cancer	1538 (1.2)	2417 (2.0)	−0.07	956 (1.4)	1011 (1.5)	−0.01
HIV/AIDS	873 (0.7)	419 (0.4)	0.04	416 (0.6)	353 (0.5)	0.01
Hemiplegia	1025 (0.8)	1587 (1.3)	−0.06	661 (1.0)	735 (1.1)	−0.01

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; DPP-4i, dipeptidyl peptidase-4 inhibitor; GLP-1 RA, glucagon-like peptide-1 receptor agonist; HbA_{1c}, hemoglobin A_{1c}; HFRS, Hospital Frailty Risk Score; SGLT2, sodium-glucose cotransporter 2; SMD, standardized mean difference.

[a] Values less than 0.10 indicate adequate balance.

[b] To convert to millimoles per mole, subtract 2.15 from HbA_{1c} percentage, then multiply by 10.929.

[c] The Charlson Comorbidity Index is a weighted sum of comorbid conditions (individual weights of 1, 2, 3, or 6). Higher scores indicate greater comorbidity burden and higher estimated mortality risk. Scores in this cohort ranged from 1 to 20.

pathway produced a small net increase in fracture risk (HR, 1.01 [95% CI, 1.00-1.02]) (**Figure 3**). The E-value was 1.83 for the intention-to-treat point estimate and 1.70 for the upper confidence bound.

Stratification by T2D Status

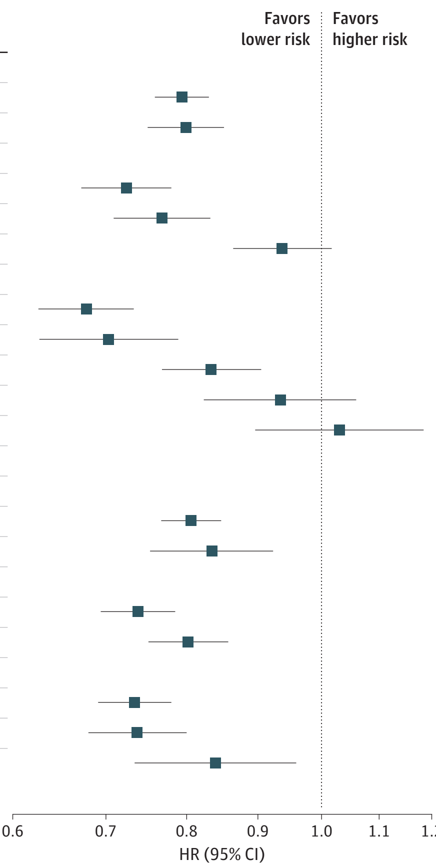
In PSM cohorts stratified by T2D status (245 365 pairs overall; 208 647 [85.0%] with T2D; 36 718 [15.0%] without T2D), GLP-1 RA use was associated with a lower fracture risk at 1 year among initiators with T2D (HR, 0.78 [95% CI, 0.73-0.83]; 3909 events; number needed to treat, 588 [95% CI, 437-895]) but not among those without T2D (HR, 1.02 [95% CI, 0.89-1.17]) (interaction *P* < .001). This divergence persisted at 3 years, with continued protection in patients with T2D (HR, 0.91 [95% CI, 0.88-0.95]) and increased fracture risk among those without T2D (HR, 1.13 [95% CI, 1.04-1.23]) (interaction *P* < .001) (**Figure 4**).

Discussion

This comparative effectiveness study using a target trial emulation of 133 606 adults found that GLP-1 RA initiation was associated with a 21% lower 3-year risk of fragility fracture compared with DPP-4i initiation (HR, 0.79 [95% CI, 0.76-0.83]; number needed to treat, 126 [95% CI, 101-168]). Although the absolute risk reduction at 3 years was modest (0.79%), this magnitude is clinically relevant given a baseline 3-year major osteoporotic fracture risk of approximately 3% to 4% in

Figure 2. Forest Plot of Glucagon-Like Peptide-1 Receptor Agonist (GLP-1 RA) vs Dipeptidyl Peptidase-4 Inhibitor (DPP-4i) and Fragility Fracture Risk, Primary, Site-Specific, and Subgroup Analyses

	ARR, %	NNT	HR (95% CI)	*P* value
Primary analysis				
ITT, 3 y	0.79 (0.60-0.99)	126 (101-168)	0.79 (0.76-0.83)	NA
Per protocol	NA	NA	0.80 (0.75-0.85)	NA
Annual HRs				
Year 1	NA	NA	0.72 (0.67-0.78)	
Year 2	NA	NA	0.77 (0.71-0.83)	NA
Year 3	NA	NA	0.94 (0.86-1.02)	
Fracture site, 3 y				
Vertebral	NA	NA	0.68 (0.63-0.73)	
Hip or femur	NA	NA	0.70 (0.63-0.79)	
Rib	NA	NA	0.83 (0.77-0.91)	<.001[a]
Proximal humerus	NA	NA	0.93 (0.82-1.06)	
Distal radius or ulna	NA	NA	1.03 (0.90-1.18)	
Subgroup analyses				
Age, y				
50-74	0.63 (0.43-0.83)	158 (121-232)	0.81 (0.77-0.85)	.56[b]
≥75	0.67 (0.55-1.75)	150 (57-183)	0.83 (0.75-0.92)	
Sex				
Female	1.11 (0.80-1.42)	90 (70-125)	0.74 (0.69-0.79)	.07[b]
Male	0.59 (0.33-0.83)	170 (120-306)	0.80 (0.75-0.86)	
HFRS				
Low (<5)	0.80 (0.60-1.00)	125 (100-167)	0.73 (0.69-0.78)	
Intermediate (5-15)	1.33 (0.80-1.86)	75 (54-125)	0.74 (0.68-0.80)	.19[b]
High (>15)	1.16 (0.39-2.51)	86 (40-259)	0.84 (0.73-0.96)	



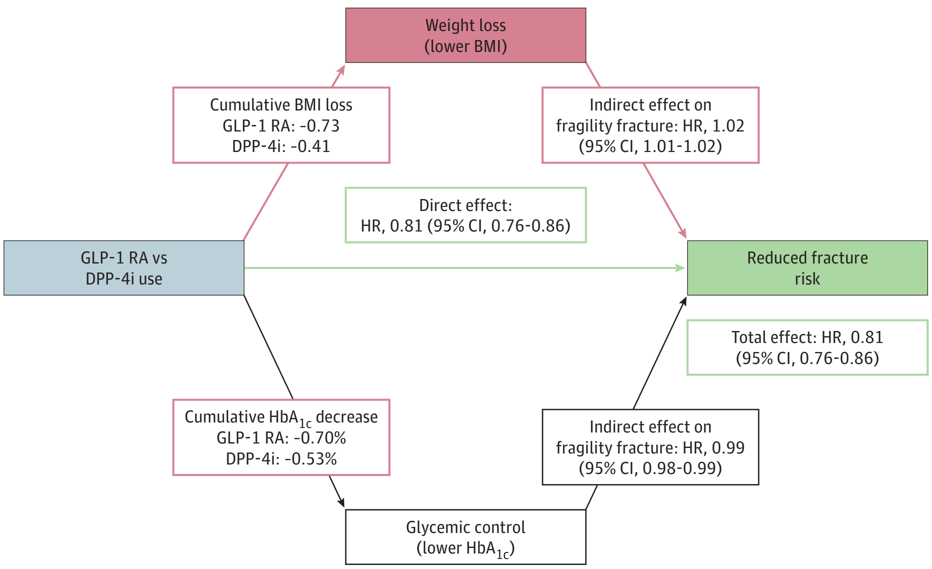
ARR indicates absolute risk reduction; HFRS, Hospital Frailty Risk Score; HR, hazard ratio; ITT, intention to treat; NA, not applicable; NNT, number needed to treat.

[a] *P* value for heterogeneity.

[b] *P* value for interaction.

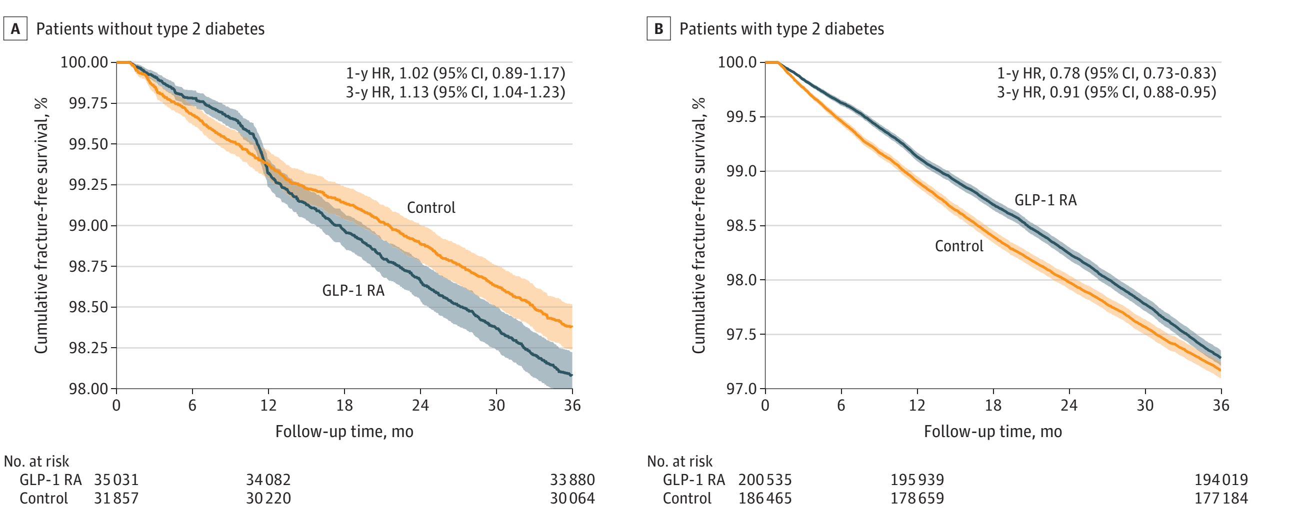
comparable populations and the substantial morbidity and mortality associated with hip and vertebral fractures.[33,34] Even small absolute reductions may translate into meaningful population-level event prevention. In landmark sensitivity analyses, protective associations were observed among adults with T2D (3-year HR, 0.91 [95% CI, 0.88-0.95]), whereas GLP-1 RA use was associated with increased fracture risk among users without T2D (HR, 1.13 [95% CI, 1.04-1.23]) (interaction $P < .001$). Risk reductions were greatest for vertebral (HR, 0.68 [95% CI, 0.63-0.73]), hip and femur (HR, 0.70 [95% CI, 0.63-0.79]), and rib fractures (HR, 0.83 [95% CI, 0.77-0.91]), with no significant associations at the distal radius or proximal humerus. Mediation analyses indicated that these associations were independent of changes in BMI and HbA_{1c}, consistent with a potential direct skeletal effect of GLP-1 RA therapy.

Figure 3. Diagram of the Mediation Analysis of Body Mass Index (BMI) and Hemoglobin A_{1c} (HbA_{1c}) Changes on the Association Between Glucagon-Like Peptide-1 Receptor Agonist (GLP-1 RA) Use and Fragility Fracture Risk



Body mass index is calculated as weight in kilograms divided by height in meters squared. DPP-4i indicates dipeptidyl peptidase-4 inhibitor; HR, hazard ratio.

Figure 4. Kaplan-Meier Curves Comparing Fragility Fracture Risk in Patients With and Without Type 2 Diabetes Mellitus After Glucagon-Like Peptide-1 Receptor Agonist (GLP-1 RA) Initiation



Control indicates patients who have never taken GLP-1 RAs. Shading indicates the 95% CI. HR indicates hazard ratio.

The persistence of fracture protection despite weight loss provides an important clinical insight. Weight loss is associated with reduced BMD and increased fracture risk, consistent with the 2% increase in fracture risk observed with the cumulative BMI loss in this study. Yet, GLP-1 RA use was associated with lower fracture risk despite these changes, suggesting that potential direct skeletal effects may outweigh the adverse consequences of weight loss.[17-19,35] Weight loss–associated bone loss may reflect skeletal unloading, as well as nutritional, endocrine, and body composition changes. Although compatible with a bone anabolic or antiresorptive effect of GLP-1 signaling, this pattern does not establish causality and may reflect unmeasured changes in physical activity, balance, or fall propensity. Accordingly, these findings do not argue against GLP-1 RA use on the basis of fracture risk, though bone health monitoring remains prudent.[24,36]

The current American Diabetes Association standards of care describe GLP-1 RAs as having neutral skeletal effects, but our findings suggest that this assumption may warrant reevaluation.[14] Our findings suggest that generalizability varies by population, with more pronounced associations in adults with T2D and more limited applicability to younger individuals or those using GLP-1 RAs exclusively for weight management. Because patients with established osteoporosis were excluded, results should not be extrapolated to individuals with known osteoporosis or prior fragility fracture. Mechanistically, incretin-based therapies support mesenchymal stem cell populations, inhibit inflammation-induced adipocyte and osteoclast differentiation, and reduce bone resorption, suggesting greater skeletal benefit in states of metabolic stress or inflammation, such as T2D.[27,37-39]

Fracture risk reduction was observed across multiple skeletal sites, with the greatest protection in the lower appendicular skeleton, pelvis, and axial skeleton and the least in the upper extremities. This distribution parallels prior evidence that GLP-1 RAs preferentially increase BMD at weight-bearing sites, such as the lumbar spine, femoral neck, and total hip, suggesting a systemic skeletal effect concentrated at regions of higher mechanical strain and turnover, consistent with Wolff law.[26,27,40-44] Notably, fractures associated with the highest morbidity and mortality, including hip and femur fractures that carry up to 12% to 36% 1-year mortality, have shown the strongest protection.[9-12] Biomarker data provide biologic plausibility for these findings. For example, meta-analyses and small clinical trials have suggested that GLP-1 RAs may increase markers of bone formation (bone-specific alkaline phosphatase, osteocalcin, and procollagen type I N-terminal propeptide) while reducing bone resorption (cross-linked C-terminal telopeptides of type I collagen), although results were heterogeneous and varied by agent.[41,45] Preclinical studies further showed activation of WNT–β-catenin signaling and increased osteoprotegerin compared with receptor activator of the nuclear factor-κB ligand, supporting reduced osteoclast-mediated resorption.[23,46] Collectively, these data support, but do not prove, a direct anabolic and antiresorptive effect of GLP-1 RAs. In this context, the observed fracture protection compares favorably with glucose-lowering therapies, such as thiazolidinediones, insulin, and sulfonylureas, which may increase fracture risk, whereas metformin and DPP-4is exhibit largely neutral skeletal profiles.[14]

The large, multicenter dataset used in this study provided substantially greater statistical power than prior investigations, exceeding the sample sizes of existing meta-analyses by more than 5-fold and individual randomized clinical trials by approximately 250-fold.[27,28,40,41] Because fragility fractures occur in only approximately 1% of older adults each year, most randomized clinical trials are underpowered to detect meaningful fracture differences (number needed to treat, 126 [95% CI, 101-168]).

Limitations

As with all observational target trial emulations using aggregated, federated, electronic health record data, several limitations warrant consideration. Treatment was not randomly assigned, and although PSM emulated key features of randomization, residual confounding remains possible, particularly from unmeasured factors such as physical activity, fall risk, frailty, muscle mass, baseline BMD, and socioeconomic factors influencing access to GLP-1 RA therapy. TriNetX does not permit access to source-level clinical records, precluding health record adjudication, validation of diagnostic coding, or assessment of site-specific heterogeneity across contributing health systems.[47] Fragility fractures occurring outside participating health systems may not have been captured. Administrative health care gaps were low in both groups (4.7% for DPP-4i and 3.3% for GLP-1 RAs), but differential follow-up or site-level variation could introduce residual confounding and may influence effect estimates in either direction. Data harmonization and administrative coding may also introduce exposure or outcome misclassification, although validated *ICD-10-CM* definitions and deidentification thresholds mitigated this risk.[47] The mediation analysis required complete longitudinal BMI and HbA_{1c} data and may therefore preferentially select for more closely monitored patients. Despite the large sample size and statistical precision, systematic bias cannot be excluded. Although fracture risk reduction was observed during the first 2 years of GLP-1 RA therapy, attenuation by year 3 may reflect declining medication adherence, delayed skeletal consequences of weight loss, or both, consistent with observations from bariatric surgery populations in which adverse skeletal effects may emerge over several years.[48,49] The E-value of 1.83 indicated moderate robustness to unmeasured confounding. Finally, as a retrospective study, causal inference cannot be established. Larger prospective and mechanistic studies are needed to clarify long-term skeletal effects across populations and potential perioperative implications of GLP-1 RAs.[50]

Conclusions

This comparative effectiveness study using target trial emulation of 133 606 adults with T2D found that GLP-1 RA initiation was associated with lower fragility fracture risk compared with DPP-4i initiation, independent of changes in BMI and HbA_{1c}. The attenuation of the association by year 3 underscores the need for continued monitoring of bone health and longer-term studies to determine whether weight loss–related skeletal effects emerge with sustained treatment. Because weight loss was associated with increased fracture risk in mediation analysis, baseline bone density assessment and monitoring during treatment may be warranted. These results challenge the assumption that GLP-1 RAs have a net neutral effect on bone. Importantly, to our knowledge, this study used the largest dataset to date to evaluate GLP-1 RAs and fracture risk. As GLP-1 RA use expands, the favorable skeletal associations observed here warrant further investigation, including in patients with osteopenia or early osteoporosis. Larger prospective randomized clinical trials and preclinical studies are needed to define the durability, magnitude, and mechanisms of these associations.

ARTICLE INFORMATION

Accepted for Publication: May 5, 2026.

Published: July 24, 2026. doi:10.1001/jamanetworkopen.2026.25141

Corresponding Author: Christopher D. Hamad, MD, Department of Orthopaedic Surgery, University of California, Los Angeles, 1225 15th St, Ste 3144B, Santa Monica, CA 90404 (chamad@mednet.ucla.edu).

Author Affiliations: Department of Orthopaedic Surgery, University of California, Los Angeles (Hamad, Wiener, Adams, Bernthal, Sheppard); Case Western Reserve University School of Medicine, Cleveland, Ohio (Wiener); Department of Population and Quantitative Health Sciences, Case Western Reserve University, Cleveland, Ohio (Wiener, Kaelber); Center for Clinical Informatics Research and Education, The MetroHealth System, Cleveland, Ohio (Wiener, Kaelber); David Geffen School of Medicine, University of California, Los Angeles (Golzar); University of Massachusetts Chan Medical School, Worcester (Kaur); The University of Texas Medical Branch Galveston School of Medicine, Galveston (Henein); The University of California, Los Angeles (Kittredge, Su); Division of Infectious Diseases, Department of Medicine, Harbor-UCLA Medical Center, Research and Education Institute at Harbor-UCLA, Torrance, California (Chan, Yeaman).

Author Contributions: Mr Wiener and Dr Sheppard had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Hamad, Wiener, Kittredge, Kaelber, Yeaman, Adams, Bernthal, Sheppard.

Acquisition, analysis, or interpretation of data: Hamad, Wiener, Golzar, Kaur, Henein, Su, Chan, Yeaman, Adams, Sheppard.

Drafting of the manuscript: Hamad, Wiener, Kaur, Henein, Kittredge, Adams, Sheppard.

Critical review of the manuscript for important intellectual content: Hamad, Wiener, Golzar, Kittredge, Su, Kaelber, Chan, Yeaman, Adams, Bernthal, Sheppard.

Statistical analysis: Hamad, Wiener, Golzar, Kittredge, Yeaman.

Obtained funding: Kaelber.

Administrative, technical, or material support: Hamad, Kaur, Henein, Kaelber.

Supervision: Hamad, Kaelber, Yeaman, Adams, Bernthal, Sheppard.

Conflict of Interest Disclosures: Dr Hamad reported receiving grants from the Musculoskeletal Tumor Society, Orthopaedic Trauma Association, H & H Lee Charitable Foundation, and San Diego Spine Foundation outside the submitted work. Dr Kaelber reported receiving grants from the National Center for Advancing Translation Sciences during the conduct of the study and being a member of the customer advisory board (unpaid) with TriNetX. Dr Bernthal reported being a consultant for Zimmer Biomet, Deciphera, and Daiichi Sankyo and receiving grants from NIH outside the submitted work. No other disclosures were reported.

Funding/Support: This study was partially funded by National Research Service Award 2T32AR059033 to the University of California, Los Angeles Regenerative Musculoskeletal Medicine Training Program from the NIH National Institute of Arthritis and Musculoskeletal and Skin Diseases (Dr Hamad). This study was also partially supported by the Clinical and Translational Science Collaborative of Northern Ohio through grant UM1TR004528 from the NIH National Center for Advancing Translational Sciences.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Disclaimer: The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

Data Sharing Statement: See Supplement 2.

Additional Information: ChatGPT 5.4 Thinking and ChatGPT 5.5 Thinking (OpenAI) were used between March 16 and May 27, 2026, to assist with language editing, formatting, and refinement of the manuscript. The authors take responsibility for the integrity of the content generated.

REFERENCES

1. Emmerich SD, Fryar CD, Stierman B, Ogden CL. Obesity and severe obesity prevalence in adults: United States, August 2021-August 2023. National Center for Health Statistics; 2024. Accessed November 6, 2025. https://www.cdc.gov/nchs/data/databriefs/db508.pdf

2. Ng M, Dai X, Cogen RM, et al; GBD 2021 US Obesity Forecasting Collaborators. National-level and state-level prevalence of overweight and obesity among children, adolescents, and adults in the USA, 1990-2021, and forecasts up to 2050. *Lancet*. 2024;404(10469):2278-2298. doi:10.1016/S0140-6736(24)01548-4

3. Wang L, Li X, Wang Z, et al. Trends in prevalence of diabetes and control of risk factors in diabetes among US adults, 1999-2018. *JAMA*. 2021;326(8):1-13. doi:10.1001/jama.2021.9883

4. Alhiary R, Kesselheim AS, Gabriele S, Beall RF, Tu SS, Feldman WB. Patents and regulatory exclusivities on GLP-1 receptor agonists. *JAMA*. 2023;330(7):650-657. doi:10.1001/jama.2023.13872

5. Sheth K, Kim S, Porterfield L, Virani SS, Wadhwani S, Vaughan EM. The expanding scope of GLP-1 receptor agonists: six uses beyond diabetes. *Curr Atheroscler Rep*. 2025;27(1):76. doi:10.1007/s11883-025-01319-6

6. Perkovic V, Tuttle KR, Rossing P, et al; FLOW Trial Committees and Investigators. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med*. 2024;391(2):109-121. doi:10.1056/NEJMoa2403347

7. Vahratian A, Warren A. GLP-1 injectable use among adults with diagnosed diabetes: United States, 2024. *NCHS Data Brief*. 2025;(537):1-8. doi:10.15620/cdc/174616

8. Montero A, Sparks G, Presiado M, Hamel L. KFF health tracking poll May 2024: the public's use and views of GLP-1 Drugs. Kaiser Family Foundation (KFF). 2024. Accessed November 6, 2025. https://www.kff.org/health-costs/kff-health-tracking-poll-may-2024-the-publics-use-and-views-of-glp-1-drugs/

9. Elmaleh-Sachs A, Schwartz JL, Bramante CT, Nicklas JM, Gudzune KA, Jay M. Obesity management in adults: a review. *JAMA*. 2023;330(20):2000-2015. doi:10.1001/jama.2023.19897

10. Tran T, Bliuc D, Hansen L, et al. Persistence of excess mortality following individual nonhip fractures: a relative survival analysis. *J Clin Endocrinol Metab*. 2018;103(9):3205-3214. doi:10.1210/jc.2017-02656

11. Walter N, Szymski D, Kurtz SM, et al. Femoral shaft fractures in elderly patients - an epidemiological risk analysis of incidence, mortality and complications. *Injury*. 2023;54(7):110822. doi:10.1016/j.injury.2023.05.053

12. Sing CW, Lin TC, Bartholomew S, et al. Global epidemiology of hip fractures: secular trends in incidence rate, post-fracture treatment, and all-cause mortality. *J Bone Miner Res*. 2023;38(8):1064-1075. doi:10.1002/jbmr.4821

13. Nicholson WK, Silverstein M, Wong JB, et al; US Preventive Services Task Force. Screening for osteoporosis to prevent fractures: US Preventive Services Task Force recommendation statement. *JAMA*. 2025;333(6):498-508. doi:10.1001/jama.2024.27154

14. American Diabetes Association Professional Practice Committee for Diabetes. 4. Comprehensive medical evaluation and assessment of comorbidities: standards of care in diabetes—2026. *Diabetes Care*. 2026;49(suppl 1):S61-S88. doi:10.2337/dc26-S004

15. Wang H, Ba Y, Xing Q, Du JL. Diabetes mellitus and the risk of fractures at specific sites: a meta-analysis. *BMJ Open*. 2019;9(1):e024067. doi:10.1136/bmjopen-2018-024067

16. Koromani F, Oei L, Shevroja E, et al. Vertebral fractures in individuals with type 2 diabetes: more than skeletal complications alone. *Diabetes Care*. 2020;43(1):137-144. doi:10.2337/dc19-0925

17. Harvey NC, Johansson H, McCloskey EV, et al. Body mass index and subsequent fracture risk: a meta-analysis to update FRAX. *J Bone Miner Res*. 2025;40(10):1144-1155. doi:10.1093/jbmr/zjaf091

18. Lv QB, Fu X, Jin HM, et al. The relationship between weight change and risk of hip fracture: meta-analysis of prospective studies. *Sci Rep*. 2015;5(1):16030. doi:10.1038/srep16030

19. Komorita Y, Iwase M, Fujii H, et al. Impact of body weight loss from maximum weight on fragility bone fractures in Japanese patients with type 2 diabetes: the Fukuoka Diabetes Registry. *Diabetes Care*. 2018;41(5):1061-1067. doi:10.2337/dc17-2004

20. Pi-Sunyer X, Astrup A, Fujioka K, et al. A randomized, controlled trial of 3.0 mg of liraglutide in weight management. *N Engl J Med*. 2015;373(1):11-22. doi:10.1056/NEJMoa1411892

21. Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med*. 2022;387(3):205-216. doi:10.1056/NEJMoa2206038

22. Johnson KC, Bray GA, Cheskin LJ, et al. The effect of intentional weight loss on fracture risk in persons with diabetes: results from the Look AHEAD randomized clinical trial. *J Bone Miner Res*. 2017;32(11):2278-2287. doi:10.1002/jbmr.3214

23. Luo G, Liu H, Lu H. Glucagon-like peptide-1(GLP-1) receptor agonists: potential to reduce fracture risk in diabetic patients? *Br J Clin Pharmacol*. 2016;81(1):78-88. doi:10.1111/bcp.12777

24. Abo-Elenin MHH, Kamel R, Nofal S, Ahmed AAE. The crucial role of beta-catenin in the osteoprotective effect of semaglutide in an ovariectomized rat model of osteoporosis. *Naunyn Schmiedebergs Arch Pharmacol*. 2025;398(3):2677-2693. doi:10.1007/s00210-024-03378-z

25. American Diabetes Association Professional Practice Committee. 4. Comprehensive medical evaluation and assessment of comorbidities: standards of care in diabetes—2025. *Diabetes Care*. 2025;48(suppl 1):S59-S85. doi:10.2337/dc25-S004

26. Tan Y, Liu S, Tang Q. Effect of GLP-1 receptor agonists on bone mineral density, bone metabolism markers, and fracture risk in type 2 diabetes: a systematic review and meta-analysis. *Acta Diabetol*. 2025;62(5):589-606. doi:10.1007/s00592-025-02468-5

27. Alalwani YJ, Alshahrani AS, Alsudays AI, et al. Differential effects of GLP-1 receptor agonists on diabetic osteopathy in type 2 diabetes: a patient-stratified network meta-analysis. *BMC Musculoskelet Disord*. 2025;26(1):742. doi:10.1186/s12891-025-09022-y

28. Zhang YS, Weng WY, Xie BC, et al. Glucagon-like peptide-1 receptor agonists and fracture risk: a network meta-analysis of randomized clinical trials. *Osteoporos Int*. 2018;29(12):2639-2644. doi:10.1007/s00198-018-4649-8

29. Zhang Y, Chen G, Wang W, Yang D, Zhu D, Jing Y. Association of glucagon-like peptide-1 receptor agonists use with fracture risk in type 2 diabetes: a meta-analysis of randomized controlled trials. *Bone*. 2025;192:117338. doi:10.1016/j.bone.2024.117338

30. Ghaferi AA, Schwartz TA, Pawlik TM. STROBE reporting guidelines for observational studies. *JAMA Surg*. 2021;156(6):577-578. doi:10.1001/jamasurg.2021.0528

31. Cashin AG, Hansford HJ, Hernán MA, et al. Transparent Reporting of Observational Studies Emulating a Target Trial-the TARGET statement. *JAMA*. 2025;334(12):1084-1093. doi:10.1001/jama.2025.13350

32. Gilbert T, Neuburger J, Kraindler J, et al. Development and validation of a hospital frailty risk score focusing on older people in acute care settings using electronic hospital records: an observational study. *Lancet*. 2018;391 (10132):1775-1782. doi:10.1016/S0140-6736(18)30668-8

33. Samelson EJ, Broe KE, Xu H, et al. Cortical and trabecular bone microarchitecture as an independent predictor of incident fracture risk in older women and men in the Bone Microarchitecture International Consortium (BoMIC): a prospective study. *Lancet Diabetes Endocrinol*. 2019;7(1):34-43. doi:10.1016/S2213-8587(18)30308-5

34. Tran T, Bliuc D, Pham HM, et al; CaMos Research Group. A risk assessment tool for predicting fragility fractures and mortality in the elderly. *J Bone Miner Res*. 2020;35(10):1923-1934. doi:10.1002/jbmr.4100

35. Zibellini J, Seimon RV, Lee CM, et al. Does diet-induced weight loss lead to bone loss in overweight or obese adults? A systematic review and meta-analysis of clinical trials. *J Bone Miner Res*. 2015;30(12):2168-2178. doi:10.1002/jbmr.2564

36. Jia IT, Bloomfield GC, Chen MY, et al. Analysis of the long-term impact of glucagon-like peptide-1 (GLP-1) receptor agonists for control of obesity and obesity-related comorbidities: a meta-analysis. *Surg Endosc*. 2025;39 (12):8580-8589. doi:10.1007/s00464-025-12086-5

37. Kitaura H, Ogawa S, Ohori F, et al. Effects of incretin-related diabetes drugs on bone formation and bone resorption. *Int J Mol Sci*. 2021;22(12):6578. doi:10.3390/ijms22126578

38. Sanz C, Vázquez P, Blázquez C, Barrio PA, Alvarez MDM, Blázquez E. Signaling and biological effects of glucagon-like peptide 1 on the differentiation of mesenchymal stem cells from human bone marrow. *Am J Physiol Endocrinol Metab*. 2010;298(3):E634-E643. doi:10.1152/ajpendo.00460.2009

39. Li Y, Fu H, Wang H, et al. GLP-1 promotes osteogenic differentiation of human ADSCs via the Wnt/GSK-3β/β-catenin pathway. *Mol Cell Endocrinol*. 2020;515:110921. doi:10.1016/j.mce.2020.110921

40. Cai TT, Li HQ, Jiang LL, et al. Effects of GLP-1 receptor agonists on bone mineral density in patients with type 2 diabetes mellitus: a 52-week clinical study. *Biomed Res Int*. Published online September 17, 2021. doi:10.1155/2021/3361309

41. Li X, Li Y, Lei C. Effects of glucagon-like peptide-1 receptor agonists on bone metabolism in type 2 diabetes mellitus: a systematic review and meta-analysis. *Int J Endocrinol*. 2024;2024:1785321. doi:10.1155/2024/1785321

42. Chai S, Liu F, Yang Z, et al. Risk of fracture with dipeptidyl peptidase-4 inhibitors, glucagon-like peptide-1 receptor agonists, or sodium-glucose cotransporter-2 inhibitors in patients with type 2 diabetes mellitus: a systematic review and network meta-analysis combining 177 randomized controlled trials with a median follow-up of 26 weeks. *Front Pharmacol*. 2022;13:825417. doi:10.3389/fphar.2022.825417

43. Hidayat K, Du X, Shi BM. Risk of fracture with dipeptidyl peptidase-4 inhibitors, glucagon-like peptide-1 receptor agonists, or sodium-glucose cotransporter-2 inhibitors in real-world use: systematic review and meta-analysis of observational studies. *Osteoporos Int*. 2019;30(10):1923-1940. doi:10.1007/s00198-019-04968-x

44. Chen JH, Liu C, You L, Simmons CA. Boning up on Wolff's Law: mechanical regulation of the cells that make and maintain bone. *J Biomech*. 2010;43(1):108-118. doi:10.1016/j.jbiomech.2009.09.016

45. Hansen MS, Wölfel EM, Jeromdesella S, et al. Once-weekly semaglutide versus placebo in adults with increased fracture risk: a randomised, double-blinded, two-centre, phase 2 trial. *EClinicalMedicine*. 2024;72: 102624. doi:10.1016/j.eclinm.2024.102624

46. Xie B, Chen S, Xu Y, et al. The impact of glucagon-like peptide 1 receptor agonists on bone metabolism and its possible mechanisms in osteoporosis treatment. *Front Pharmacol*. 2021;12:697442. doi:10.3389/fphar.2021.697442

47. Goldstein ND, Olivieri-Mui B, Burstyn I. Are aggregated electronic health record datasets good for research? *J Gen Intern Med*. 2025;40(15):3743-3749. doi:10.1007/s11606-025-09808-9

48. Frenkel Rutenberg T, Rutenberg R, Kosashvili Y, Warschawski Y, Iordache SD. Bariatric surgery and the risk for osteoporotic fractures - a population-based study. *Arch Orthop Trauma Surg*. 2025;145(1):299. doi:10.1007/s00402-025-05899-5

49. Robinson DE, Douglas I, Tan GD, et al. Bariatric surgery increases the rate of major fracture: self-controlled case series study in UK Clinical Practice Research Datalink. *J Bone Miner Res*. 2021;36(11):2153-2161. doi:10.1002/jbmr.4405

50. Wiener JM, Sanghvi PA, Vlastaris K, et al. Glucagon-like peptide-1 receptor agonist medications alter outcomes of spine surgery: a study among over 15,000 patients. *Spine (Phila Pa 1976)*. 2025;50(13):871-880. doi:10.1097/BRS.0000000000005283

SUPPLEMENT 1.
eTable 1. Code Reference Guide

eTable 2. Baseline Demographics and Clinical Characteristics After Propensity Score Matching Among Patients With and Without Diabetes
eTable 3. Usage of GLP-1 RA Medications Within the Cohort
eTable 4. Follow-Up Duration, Censoring Patterns, and Treatment Persistence in the Propensity Score–Matched Cohort
eFigure. Sensitivity Analyses for Primary Results

SUPPLEMENT 2.
Data Sharing Statement